

Maatram Engineering Training Guide:

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Life Skills:

Technical Skills:

Foundation Skills: Computer Science 101

- The nature of computers and code, what they can and cannot do
- How computer hardware works: chips, cpu, memory, disk
- Necessary jargon: bits, bytes, megabytes, gigabytes
- How software works: what is a program, what is "running"
- How digital images work
- Computer code: loops and logic
- Big ideas: abstraction, logic, bugs
- How structured data works
- How the internet works: ip address, routing, ethernet, wi-fi
- Computer security: viruses, trojans, and passwords, oh my!
- Analog vs. digital
- Digital media, images, sounds, video, compression

Course Link: <https://online.stanford.edu/courses/soe-yccscs101-computer-science-101>

Compilers:

This self-paced course will discuss the major ideas used today in the implementation of programming language compilers, including lexical analysis, parsing, syntax-directed translation, abstract syntax trees, types and type checking, intermediate languages, dataflow analysis, program optimization, code generation, and runtime systems. As a result, you will learn how a program written in a high-level language designed for humans is systematically translated into a program written in low-level assembly more suited to machines. Along the way we will also touch on how programming languages are designed, programming language semantics, and why there are so many kinds of programming languages.

Course Link: <https://online.stanford.edu/courses/soe-yccscs1-compilers>

Introduction to Computer Science:

- A broad and robust understanding of computer science and programming

- How to think algorithmically and solve programming problems efficiently.
- Concepts like abstraction, algorithms, data structures, encapsulation, resource management, security, software engineering, and web development
- Familiarity in several languages, including C, Python, SQL, and JavaScript plus CSS and HTML
- How to engage with a vibrant community of like-minded learners from all levels of experience
- How to develop and present a final programming project to your peers

Course Link: <https://www.edx.org/learn/computer-science/harvard-university-cs50-s-introduction-to-computer-science>

Data Engineering Skills:

Databases: Relational Databases and SQL

This course introduces relational databases and comprehensive coverage of SQL, the long-accepted standard query language for relational database systems.

Course Link:

Stanford: <https://online.stanford.edu/courses/soe-ydatabases0005-databases-relational-databases-and-sql>

Harvard: <https://www.edx.org/learn/sql/harvard-university-cs50-s-introduction-to-databases-with-sql>

Databases: Advanced Topics in SQL

The Indexes and Transactions section of this course covers two important features of database systems from the application-builder's perspective: indexing for increased performance, and transactions for concurrency control and failure recovery.

The Constraints and Triggers section of this course explains key, referential integrity, and "check" constraints, followed by comprehensive coverage of database triggers.

The Views and Authorization section of this course provides extensive coverage of how database views can be created, used, and updated, and introduces standard techniques for authorization in relational databases.

<https://online.stanford.edu/courses/soe-ydatabases0001-databases-advanced-topics-sql>

Databases: Modeling and Theory

The Relational Algebra section of this course teaches the algebraic query language that provides the formal foundations of SQL.

The Relational Design Theory section of the course provides comprehensive coverage of dependency theory and normal forms in relational databases, a well-accepted theoretical framework for developing good relational database schemas.

The Unified Modeling Language section of this course introduces the data-modeling component of UML, and describes how UML diagrams are translated to relational database schemas.

Course link: <https://online.stanford.edu/courses/soe-ydatabases0003-databases-modeling-and-theory>

Databases: OLAP and Recursion

The On-Line Analytical Processing section of this course introduces star schemas, the data cube concept, and OLAP features in relational databases including the Cube and Rollup operators. The Recursion section of this course covers the SQL standard for queries over recursively defined relations.

<https://online.stanford.edu/courses/soe-ydatabases0002-databases-olap-and-recursion>

Machine Learning Specialization Track:

- Build ML models with NumPy & scikit-learn, build & train supervised models for prediction & binary classification tasks (linear, logistic regression)
- Build & train a neural network with TensorFlow to perform multi-class classification, & build & use decision trees & tree ensemble methods
- Apply best practices for ML development & use unsupervised learning techniques for unsupervised learning including clustering & anomaly detection
- Build recommender systems with a collaborative filtering approach & a content-based deep learning method & build a deep reinforcement learning model

<https://www.coursera.org/specializations/machine-learning-introduction>

Data Science Track with Marketing Domain Specialization:

Path to Data Science:

Statistics: <https://www.coursera.org/learn/stanford-statistics>

Technical: (Start with Excel)

- <https://www.edx.org/learn/data-analysis/microsoft-introduction-to-data-analysis-using-excel>
- <https://www.edx.org/learn/excel/microsoft-analyzing-and-visualizing-data-with-excel>

Data Analysis with Python:

<https://www.freecodecamp.org/learn/data-analysis-with-python/>

Domain (Business): Marketing Business

Introductions to Marketing:

<https://www.coursera.org/learn/wharton-marketing>

Introductions to Marketing Analytics

<https://www.coursera.org/learn/uva-darden-market-analytics>

Soft Skills: How to present the business insights?

<https://www.coursera.org/learn/data-analysis-project-pwc>

Canva: Tool to enrich the visualization.

<https://www.coursera.org/projects/how-to-create-presentations-using-canva>

1. **Domain:** <https://www.coursera.org/specializations/finance-accounting>

a. **Fundamentals of Finance:** <https://www.coursera.org/learn/finance-fundamentals?specialization=finance-accounting>

- In this course, you'll learn the fundamentals of corporate finance. Based on the pre-term qualifying courses for Wharton MBA students, Professor Jessica Wachter has designed this course for learners who need a refresher in financial concepts, or for those who are learning about corporate finance for the first time. You'll identify foundational concepts in corporate finance, such as NPV, Compound and Simple Interest, and Annuities versus Perpetuities. You'll also learn how to apply the NPV framework to calculating fixed-income valuation and Equity, using hypothetical examples of corporate projects. By the end of this course, you'll have honed your skills in calculating risk and returns to optimize investments and be able to assess the right set of financial information to achieve better returns for your firm.

b. **Introduction to Corporate Finance:** <https://www.coursera.org/learn/wharton-finance?specialization=finance-accounting>

- This course provides a brief introduction to the fundamentals of finance, emphasizing their application to a wide variety of real-world situations spanning personal finance, corporate decision-making, and financial intermediation. Key concepts and applications include: time value of money, risk-return tradeoff, cost of capital, interest rates, retirement savings, mortgage financing, auto leasing, capital budgeting, asset valuation, discounted cash flow (DCF) analysis, net present value, internal rate of return, hurdle rate, payback period.

c. **Introduction to Financial Accounting:** <https://www.coursera.org/learn/wharton-accounting?specialization=finance-accounting>

- Master the technical skills needed to analyze financial statements and disclosures for use in financial analysis, and learn how accounting standards and managerial incentives affect the financial reporting process. By the end of this course, you'll be able to read the three most common financial statements: the income statement, balance sheet, and statement of cash flows. Then you can apply these skills to a real-world business challenge as part of the Wharton Business Foundations Specialization.

d. **More Introduction to Financial Accounting:** <https://www.coursera.org/learn/wharton-financial-accounting?specialization=finance-accounting>

- The course builds on my Introduction The course builds on my Introduction to Financial Accounting course, which you should complete first. In this course, you will learn how to read, understand, and analyze most of the information provided by companies in their financial statements. These skills will help you make more informed decisions using financial information.

2. Mandatory Skills:

Microsoft Office 365: <https://www.coursera.org/specializations/microsoft-365-fundamentals?>

a. **Work Smarter with Microsoft Word:** <https://www.coursera.org/learn/microsoft-word-work-smarter?specialization=microsoft-365-fundamentals>

- How to create and maintain professional-looking reports, multi-column newsletters, résumés, and business correspondence.
- How to manage documents and document collaboration
- How to insert and format text and graphic elements
- How to manage tables and lists

b. **Work Smarter with Microsoft Excel:** <https://www.coursera.org/learn/microsoft-excel-work-smarter?specialization=microsoft-365-fundamentals>

- You will learn how to create a spreadsheet, work with formulas and functions, make your spreadsheets look professional, and print a spreadsheet.
- How to manage worksheets, workbooks, tables table data, data cells and ranges, and charts
- How to perform operations by using formulas and functions

c. **Work Smarter with Microsoft PowerPoint:**
<https://www.coursera.org/learn/microsoft-powerpoint-work-smarter?specialization=microsoft-365-fundamentals>

- How to create, save, and present a basic presentation; apply the basics of formatting; and add multimedia to a presentation
- How to manage slides and presentations
- How to apply transitions and animations
- How to insert elements into a presentation

Skills for Data Science with Marketing Business Specialization:

Statistics: <https://www.coursera.org/learn/stanford-statistics>

Domain (Business): Marketing Business

- Introductions to Marketing:
<https://www.coursera.org/learn/wharton-marketing>
- Introductions to Marketing Analytics
<https://www.coursera.org/learn/uva-darden-market-analytics>
- Soft Skills: How to present the business insights
<https://www.coursera.org/learn/data-analysis-project-pwc>

Advance Excel: <https://www.edx.org/learn/data-analysis/microsoft-introduction-to-data-analysis-using-excel>

- Create flexible data aggregations using pivot tables
- Represent data visually using pivot charts
- Calculate margins and other common ratios using calculation on pivot table
- Filter data using slicers in multiple pivot tables
- Create aggregate reports using formula-based techniques

Databases: Relational Databases and SQL - : <https://www.edx.org/learn/sql/harvard-university-cs50-s-introduction-to-databases-with-sql>

- This course introduces relational databases and comprehensive coverage of SQL, the long-accepted standard query language for relational database systems.

Skills for IT Support

#1 Computer Science 101:

<https://online.stanford.edu/courses/soe-ycscs101-computer-science-101>

- The nature of computers and code, what they can and cannot do
- How computer hardware works: chips, cpu, memory, disk
- Necessary jargon: bits, bytes, megabytes, gigabytes
- How software works: what is a program, what is "running"
- How digital images work
- Computer code: loops and logic
- Big ideas: abstraction, logic, bugs
- How structured data works

- How the internet works: ip address, routing, ethernet, wi-fi
- Computer security: viruses, trojans, and passwords, oh my!
- Analog vs. digital
- Digital media, images, sounds, video, compression

#2 Introduction to Computer Science:

<https://www.edx.org/learn/computer-science/harvard-university-cs50-s-introduction-to-computer-science>

- A broad and robust understanding of computer science and programming
- How to think algorithmically and solve programming problems efficiently.
- Concepts like abstraction, algorithms, data structures, encapsulation, resource management, security, software engineering, and web development
- Familiarity in several languages, including C, Python, SQL, and JavaScript plus CSS and HTML
- How to engage with a vibrant community of like-minded learners from all levels of experience
 - How to develop and present a final programming project to your peers

#3 Google IT Support Professional Certificate

- Gain skills required to succeed in an entry-level IT job.
- Learn to perform day-to-day IT support tasks including computer assembly, wireless networking, installing programs, and customer service.
- Learn how to provide end-to-end customer support, ranging from identifying problems to troubleshooting and debugging.
- Learn to use systems including Linux, Domain Name Systems, Command-Line Interface, and Binary Code

<https://www.coursera.org/professional-certificates/google-it-support>

Technical Support Fundamentals

This course is the first of a series that aims to prepare you for a role as an entry-level IT Support Specialist. In this course, you'll be introduced to the world of Information Technology, or IT. You'll learn about the different facets of Information Technology, like computer hardware, the Internet, computer software, troubleshooting, and customer service. This course covers a wide variety of topics in IT that are designed to give you an overview of what's to come in this certificate program.

By the end of this course, you'll be able to: • understand how the binary system works • assemble a computer from scratch • choose and install an operating system on a computer • understand what the Internet is, how it works, and the impact it has in the modern world • learn how applications are

created and how they work under the hood of a computer • utilize common problem-solving methodologies and soft skills in an Information Technology setting

<https://www.coursera.org/learn/technical-support-fundamentals>

The Bits and Bytes of Computer Networking

This course is designed to provide a full overview of computer networking. We'll cover everything from the fundamentals of modern networking technologies and protocols to an overview of the cloud to practical applications and network troubleshooting.

By the end of this course, you'll be able to: • describe computer networks in terms of a five-layer model • understand all of the standard protocols involved with TCP/IP communications • grasp powerful network troubleshooting tools and techniques • learn network services like DNS and DHCP that help make computer networks run • understand cloud computing, everything as a service, and cloud storage

<https://www.coursera.org/learn/computer-networking>

Operating Systems and You: Becoming a Power User

In this course -- through a combination of video lectures, demonstrations, and hands-on practice -- you'll learn about the main components of an operating system and how to perform critical tasks like managing software and users, and configuring hardware.

By the end of this course you'll be able to: • navigate the Windows and Linux filesystems using a graphical user interface and command line interpreter • set up users, groups, and permissions for account access • install, configure, and remove software on the Windows and Linux operating systems • configure disk partitions and filesystems • understand how system processes work and how to manage them • work with system logs and remote connection tools • utilize operating system knowledge to troubleshoot common issues in an IT Support Specialist role

<https://www.coursera.org/learn/os-power-user>

System Administration and IT Infrastructure Services

This course will transition you from working on a single computer to an entire fleet. Systems administration is the field of IT that's responsible for maintaining reliable computer systems in a multi-user environment. In this course, you'll learn about the infrastructure services that keep all organizations, big and small, up and running. We'll deep dive on cloud so that you'll understand everything from typical cloud infrastructure setups to how to manage cloud resources. You'll also learn how to manage and configure servers and how to use industry tools to manage computers,

user information, and user productivity. Finally, you'll learn how to recover your organization's IT infrastructure in the event of a disaster.

By the end of this course you'll be able to:

- utilize best practices for choosing hardware, vendors, and services for your organization
- understand how the most common infrastructure services that keep an organization running work, and how to manage infrastructure servers
- understand how to make the most of the cloud for your organization
- manage an organization's computers and users using the directory services, Active Directory, and OpenLDAP
- choose and manage the tools that your organization will use
- backup your organization's data and know how to recover your IT infrastructure in the case of a disaster
- utilize systems administration knowledge to plan and improve processes for IT environments

<https://www.coursera.org/learn/system-administration-it-infrastructure-services>

IT Security: Defense against the digital dark arts

This course covers a wide variety of IT security concepts, tools, and best practices. It introduces threats and attacks and the many ways they can show up. We'll give you some background of encryption algorithms and how they're used to safeguard data. Then, we'll dive into the three As of information security: authentication, authorization, and accounting. We'll also cover network security solutions, ranging from firewalls to Wifi encryption options. The course is rounded out by putting all these elements together into a multi-layered, in-depth security architecture, followed by recommendations on how to integrate a culture of security into your organization or team.

At the end of this course, you'll understand:

- how various encryption algorithms and techniques work as well as their benefits and limitations.
- various authentication systems and types.
- the difference between authentication and authorization.
- how to evaluate potential risks and recommend ways to reduce risk.
- best practices for securing a network.
- how to help others to grasp security concepts and protect themselves.
- new AI skills from Google experts to help complete IT tasks.

<https://www.coursera.org/learn/it-security>

#4 Microsoft IT Support Specialist Professional Certificate

- Gain skills to succeed in Information technology systems analysis and maintenance.
- Apply concepts such as Data Backup, Cloud Computing, and Mobile Device Management to practical IT scenarios.
- Demonstrate a holistic understanding of IT support, combining hardware, software, networking, and security concepts in problem-solving.
- Learn to adapt diagnostic skills to diverse IT issues within a practical business scenario.

<https://www.coursera.org/professional-certificates/microsoft-it-support-specialist>

Introduction to Computers

- Learn about computer components like CPU, memory, and storage.
- Learn how computer system components, such as the CPU, memory, and storage devices, interact with the operating system.
- Gain insight into security, hardware, and software relationships.

<https://www.coursera.org/learn/introduction-to-computers>

Introduction to Secure Networking

- Learn computer networking basics, like TCP/IP and OSI models, and how devices collaborate.
- Gain insights into network infrastructure and monitoring techniques for effective management.
- Learn about network security components, approaches, and mitigation strategies.

<https://www.coursera.org/learn/introduction-to-secure-networking>

Essential Aspects of Software, Hardware, and Data Backup

- Gain insight into key aspects of IT security and data management.
- Develop an understanding of secure software setup and implementation.
- Learn effective strategies for hardware configurations and data backup and recovery.

<https://www.coursera.org/learn/essential-aspects-of-software-hardware-and-data-backup>

Cybersecurity and Privacy

- Gain insight into the fundamentals of cybersecurity concepts for ensuring user data privacy.
- Learn how to secure digital communications, optimize network performance, and understand how VPNs work.
- Gain knowledge about the diverse landscapes of on-premises computing, cloud services, and server communication.

<https://www.coursera.org/learn/cybersecurity-and-privacy>

The Microsoft 365 Ecosystem

- Gain insight into the components and services within Microsoft 365, such as Office applications, collaboration tools, and cloud-based services.
- Understand the integration between Microsoft 365 applications and their pivotal role in enhancing business productivity and security.
- Learn about the automation tools available within Microsoft 365 and how to leverage them to streamline workflows and improve overall efficiency.

<https://www.coursera.org/learn/the-microsoft-365-ecosystem?specialization=microsoft-it-support-specialist>

Technical Diagnostics and Troubleshooting Techniques.

- Gain insight into troubleshooting approaches for both hardware and software issues across various operating systems

- Learn to apply a systematic approach to diagnose technical issues across different hardware and software environments.
- Master diagnostic tools for efficient problem-solving in system maintenance, upgrades, network connectivity, software setup, and security

<https://www.coursera.org/learn/technical-diagnostics-and-troubleshooting-techniques?specialization=microsoft-it-support-specialist>

Introduction In User Experience Design

#1 Google UX Design Professional Certificate

- Follow the design process: empathize with users, define pain points, ideate solutions, create wireframes and prototypes, test and iterate on designs.
- Understand the basics of UX research, like planning research studies, conducting interviews and usability studies, and synthesizing research results.
- Apply foundational UX concepts, like user-centered design, accessibility, and equity-focused design.
- Create a professional UX portfolio, including end-to-end projects, so that you're ready to apply for jobs.

<https://www.coursera.org/professional-certificates/google-ux-design>

Foundations of User Experience (UX) Design

- Identify common job responsibilities of entry-level UX designers and other teams you might work with.
- Understand foundational concepts in UX design, such as user-centered design, the design process, accessibility, and equity-focused design.
- Explain why design sprints are an important and useful part of a UX designer's work.

<https://www.coursera.org/learn/foundations-user-experience-design>

Start the UX Design Process: Empathize, Define, and Ideate

- Empathize with users to understand their needs and pain points.
- Develop problem statements to define user needs.
- Generate ideas for possible solutions to user problems.

<https://www.coursera.org/learn/start-ux-design-process>

Build Wireframes and Low-Fidelity Prototypes

- Create storyboards to come up with ideas about solutions to user needs.
- Create wireframes on paper and digitally in the design tool Figma.
- Build paper prototypes to create interactive designs.
- Design low-fidelity prototypes in Figma.

<https://www.coursera.org/learn/wireframes-low-fidelity-prototypes?specialization=google-ux-design>

Conduct UX Research and Test Early Concepts

- Plan and conduct moderated and unmoderated usability studies.
- Synthesize observations from usability studies and come up with insights.
- Share research methodology and insights using persuasive presentation skills.
- Modify low-fidelity designs based on research insights.

<https://www.coursera.org/learn/conduct-ux-research?specialization=google-ux-design>

Create High-Fidelity Designs and Prototypes in Figma

- Build mockups and high-fidelity prototypes in the design tool Figma.
- Define and apply common visual design elements and principles.
- Demonstrate how design systems can be used to organize, standardize, and enhance designs.
- Understand the role of design critique sessions and feedback while iterating on designs.

<https://www.coursera.org/learn/high-fidelity-designs-prototype?specialization=google-ux-design>

Build Dynamic User Interfaces (UI) for Websites

- Apply each step of the UX design thinking framework (empathize, define, ideate, prototype, test) to create a dynamic website.
- Plan information architecture and sitemaps for website designs.
- Apply common layouts for web pages.
- Complete a design project and include it in your professional UX portfolio.

<https://www.coursera.org/learn/responsive-web-design-adobe-xd?specialization=google-ux-design>

Design a User Experience for Social Good & Prepare for Jobs

- Apply each step of the UX design thinking framework (empathize, define, ideate, prototype, test) to create a project focused on social good.
- Build wireframes, mockups, and low-fidelity and high-fidelity prototypes for a dedicated mobile app and a responsive website.
- Prepare to successfully interview for an entry-level UX design job.
- Gain a competitive edge by learning AI skills from Google experts.

<https://www.coursera.org/learn/ux-design-jobs>

<https://www.udacity.com/course/product-design--ud509>

Learning Basic Design Principles

<https://lawsofux.com/>

<https://blog.uxtweak.com/ux-laws-and-principles/>

<https://atomicdesign.bradfrost.com/chapter-2/>

<https://developer.apple.com/design/human-interface-guidelines>

Investing time in Design tools - References

<https://youtu.be/g6rQFP9zCAM?si=DBOZUsnMQpqiWW9M>

(Learn more about other design skills in <https://www.youtube.com/@envatotuts/playlists>

<https://www.youtube.com/@envatotuts>)

<https://www.youtube.com/@DesignPilot/playlists>

Quality Assurance Career Path

#1 Introduction to Testing

- Why QA and its importance in Software Development
- Understanding Software Development Life Cycle (SDLC) and Testing Life Cycle (STLC)

#2 Testing Fundamentals

- Test Case design
- Test planning and strategy
- Testing Stages/Levels: Unit, Integration, System, Acceptance
- "Types of Testing: Functional vs Non-functional testing, Manual vs Automation"

#3 Manual Testing Types

- Black-box testing
- White-box testing
- Gray-box testing
- Manual Testing Techniques
 - Boundary value analysis

- Equivalence partitioning
- Other Types of Testing
 - Exploratory Testing
 - Smoke Testing
 - Regression Testing

#4 Tools for Manual Testing

- Bug reporting and tracking tools (eg., JIRA, Bugzilla, Azure Devops)

#5 Introduction to Automation Testing

- Benefits of automation
- Overview of automation testing tools (eg., Selenium, Playwright, Cypress)
- Knowledge on any scripting languages (Python, Java, JavaScript, C#)

#6 Selenium Automation Testing

- Creating and Running Automated Tests

#7 API Testing

- Understanding APIs
- Tools for API Testing: Postman, SoapUI or Rest Assured
- API Testing with Postman (Writing and executing simple API tests)

#8 Performance Testing

- Understanding Load and Stress Testing
- Tools and techniques: JMeter, Load Runner

#9 Security Testing

- Understanding Security Testing

#10 Data Testing: SQL

- Writing and executing basic sql queries

#11 Other Skills for QA professionals

- Analytical Skill
- Attention to Detail
- Communication skills
- Logical Thinking

QA Learning Path Reference

<https://www.freecodecamp.org/learn/quality-assurance/#quality-assurance-projects>

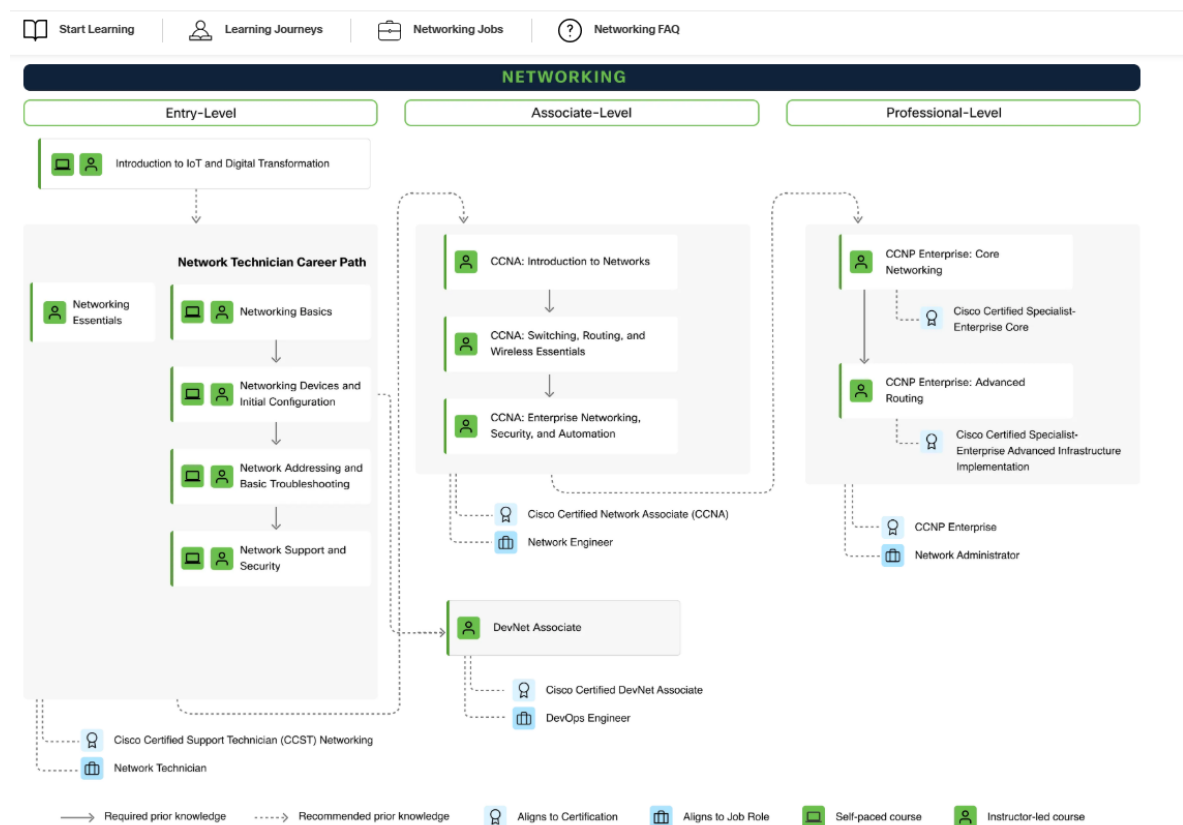
<https://www.coursera.org/specializations/software-testing-automation>

- Distinguish between verification and validation describing the key differences between them.
- Write automated functional tests for both front-end and back-end code.
- Measure the fault-finding effectiveness of a functional test suite using mutation testing.

Defend program correctness through the use of formal methods, specifically proof obligations and model checking.

Network Engineering:

Refer <https://www.netacad.com/>



Network Technician

Networks keep the digital world connected. Learn how networks work and gain employable skills to start your tech career.

Networking technology powers the Internet. Networks make it possible for us to use social media, streaming services like Netflix, ride-hailing services like Uber, online shopping through Amazon, and much more. Put simply, networks are critical for every industry, which is why demand for networking skills keeps rising. A strong foundational knowledge of networks can lead to a successful career in IT.

The Network Technician Career Path prepares you for the entry-level [Cisco Certified Support Technician \(CCST\) Network](#) certification and entry-level roles such as Network Support Technician, Entry-level Help Desk Technician, or IT Support Specialist. You will learn how networks operate, including the devices, media, and protocols that enable network communication. Develop key skills so you can perform basic troubleshooting, using effective methodologies and help desk best practices.

1. Networking Basics
2. Networking Devices and Initial Configuration
3. Network addressing and basic troubleshooting.
4. Network support and security
5. Network Technician Career Path Certification
6. Cisco Certified Support Technician Networking

Junior Cybersecurity Analyst:

<https://www.netacad.com/career-paths/cybersecurity?courseLang=en-US>

Cyber attacks increase with every digital connection made around the world. Learn how to protect and defend an organization and gain employable skills to start your cyber career.

Cyber threats affect us all, and cybersecurity jobs continue to grow. In fact, cybersecurity failure is among the top 5 global risks according to the World Economic Forum Global Risks Report (2021). The world needs diverse, tech-savvy cyber professionals to defend against shrewd digital adversaries.

This Junior Cybersecurity Analyst Career Path prepares you for the entry-level [Cisco Certified Support Technician \(CCST\) Cybersecurity](#) certification and entry-level cybersecurity positions such as Cybersecurity Technician, Junior Cybersecurity Analyst, or Tier 1 Help Desk Support roles. You will learn vital skills like threat intelligence, network security, and risk management to protect yourself and organizations from cyber attacks. If you enjoy solving puzzles, are passionate about technology, and like working in teams, the field of cybersecurity may be a great fit for you!

1. Introduction to Basics
2. Networking Basics
3. Networking Devices and Configuration
4. Endpoint security
5. Network Defense

6. Cyber Threat Management
7. Junior Cybersecurity Analyst Career path exam
8. Cisco certified support technician cybersecurity

IT Support Specialist Career Path

Become a certified IT Support Specialist! Learn to troubleshoot, help users, and secure systems. Your path to entry-level IT roles starts here.

Entry-level IT support roles, such as IT End User Support, Help Desk Support, and Computer Support Specialist, are great starting points that offer you a path into the tech industry.

The IT Support Specialist Career Path is designed to prepare you for the [Cisco Certified Support Technician \(CCST\) IT Support certification](#) and entry-level jobs. You will learn practical IT skills like managing support tickets, documenting solutions, and communicating effectively with users. You will also learn to tackle a wide range of problems, from troubleshooting hardware and resolving connectivity issues, to navigating different operating systems and applications. Finally, you will explore cloud technology basics and learn to identify and combat common security threats.

1. IT Customer Support Basics
2. Operating Systems Support
3. Security and Connectivity Support
4. Hardware and upgrade Support
5. Cisco Certified Support Technician

